CHAPTER

19

Pharmaceutical excipients: special focus on adverse interactions

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19.1 Introduction

In general, children need very few medications compared to adults to treat ailments. Thus the impetus to design and to develop the pediatric formulation is very slow. In the current clinical setup, the medications given to the pediatric population belong to the adult medicines, and the labeling information for dosing in the pediatric population is missing. These medications are often not appropriate for pediatric patients, thereby causes side effects (Whittaker et al., 2009).

The most common side effects observed in the pediatric population due to the toxic excipients used in pediatric dosage forms include bronchospasm due to the benzalkonium chloride. Similarly, aspartame use in the pediatric dosage forms leads to headaches and seizures. Propylene glycol used in the pediatric population leads to hyperosmolarity and lactic acidosis (Fabiano et al., 2011). The US government started evaluating the clinical effects of these excipients on the pediatric population. However, both long-term and short-term clinical effects of these excipients used in pediatrics are not yet clearly understood (Food and Drug Administration, 1998). Therefore the main goal of this chapter is to provide a holistic picture of the pharmaceutical excipient quality and safety aspects used in off-labeled pediatric formulations.

19.1.1 History of the pharmaceutical excipients

In contrast to conventional medicine systems like Ayurveda, Homeopathy, Unani, etc., the requirement of instant therapeutic action in critical medical conditions gave birth to the world of Allopathy. The 19th century witnessed various extremely potent synthetic therapeutic molecules, which we generally call active pharmaceutical ingredients (APIs). To administer them in an appropriate dose, these APIs need a suitable dosage form. Apart from API, there is a significant requirement of some ingredients that could assist in developing an ideal dosage form, and from there, the evolution of excipients began. Ancient practices of using honey, tree saps, and alcoholic beverages in making medicines transformed into a more determined and strapping utilization of chemicals as excipients. Various devastating excipients were discovered during the late 19th century, which meticulously revolutionized pharmaceuticals (Roy, 2015; Haywood and Glass, 2011).

Along with significant achievements, the revolutionary era of developments in pharmaceuticals brought some tragedies also. On the one hand, where progress in medical sciences helped patients surmount illness, ingestion of adulterated and contaminated ingredients in medicines sacrificed their lives. To resolve this issue, a committee was constituted in 1991, the "International Pharmaceutical Excipients Council (IPEC)," which was firmly focused on protecting the lives of individuals around the globe for over a quarter of a century. IPEC works over four primary goals, that is, (1) harmonization of incomparable standards for excipients, (2) formation of standardized "good manufacturing practices (GMP)" guidelines to assure excipients quality and safety, (3) harmonization of standard safety assessment guidelines for new excipients, and (4) rotation of productive communication between manufacturers, users, pharmacopeia officials, and regulatory authorities. GMP standards for bulk pharmaceutical excipients were jointly adopted, published, and implemented for the first time by combined efforts of IPEC-America and IPEC-Europe in 1995. To date, more than 20 guideline documents have been produced by IPEC on quality assurance and updated standards for the manufacturing of excipients for the pharmaceutical purpose, which is now being widely used on a global basis.

19.2 What are pharmaceutical excipients?

The word excipient is acquired from the Latin word "*excipere*," meaning "to except,"-which is simply explained as "other than". The standard definition of excipients presented by IPEC is "excipients are any substance other than active drug or prodrug that has been appropriately evaluated for safety and is included in a drug delivery system" for any of the following purposes (Veeravalli et al., 2020):

- Assist in the processing of the formulation during manufacture.
- Safeguarding, support, and enhancement in stability and bioavailability.
- Aids in the identification of products.
- Improvement of other aspects for overall safety and efficiency of the drug product during storage or use.

The significance of excipients in medicinal products is proved by the expression frequently used to depict them as "pharmaceutical necessities." Overall, excipients are the materials that play a vital role in fabricating various dosage forms, resolving global issues like solubility of synthetic APIs, and ultimately enhancing their bioavailability (Katdare and Chaubal, 2006).

19.2.1 The pharmaceutical perspective of excipients

Excipients play a multifaceted role in pharmaceuticals, whether it is a matter of imparting palatability to oral medication or minimizing pain and irritation on parenteral administration excipients have emerged as a potential solution (Aleeva et al., 2009). Over the past three decades, excipients have gained remarkable attraction in functionalizing and impacting the finished product's performance. The revolutionary emergence of novel dosage forms, combination products, and technologies in pharmaceuticals has substantially increased the demand for excipients. Various roles of excipients can be categorized according to the route of administration of finished pharmaceutical products (Table 19.1).

In the current scenario, thousands of excipients are being utilized in the pharmaceuticals and make up on an average of 90% of each finished product. Excipients represent a current market value of almost $5 billion, accounting for 0.6% of the total pharmaceutical market.

19.2.2 Emergence and fundamental requirement of excipients

The ground-breaking progress in the development of novel formulations demands newer and more potential excipients. It is very much understood that the quantity of excipient (s) is much more than API in a dosage form. Since excipient plays a significant role in pharmaceutical formulations, it is difficult for formulation scientists to opt and comprehend the different types/grades of excipients available. Furthermore, the cost, availability, and biopharmaceutical perspective of the selected excipient are also very critical. Hence, it has become more essential for the evolution of new excipients that are attuned not only with contemporary processing and production machinery (revolving granulators, sophisticated compactors, etc.) but also with novel active principles coming from the latest biotechnological fields and modern peptide synthesis.

The fundamental requirements for excipients rely on certain aspects that need to be critically considered. The primary requirement of various types of excipients consists of their quality, safety, and functionality. Due to an increase in counterfeit medications and the gradual loss of patent protection, regulatory organizations have created regulatory layers to assure the beneficial impact of excipients on the final products' safety, quality, efficacy, functionality, and stability. Excipients' key problems in terms of safety and function requirements are toxicity and bioavailability. Excipients can interact physically with the active component, such as nitrazepam degradation due to colloidal silica adsorption, or chemically, such as the incompatibility of hydrogen-donating medications like lansoprazole and famotidine with polyvinylpyrrolidone (PVP).

19.2.3 Roles of excipients

In previous days, it was considered that excipients were inert. But they are primarily used to increase the stability, release, and absorption of APIs. On the other hand, a major type of excipients is generally used to increase the solubility of poorly soluble APIs. Others are used as a disintegrant, binder, lubricant, filler agent, and sweetener to make medicine easier and give patients a pleasing flavor. In fact, excipients can have a beneficial or harmful impact on formulations, particularly inside live cells (Abdellah et al., 2015).

TABLE 19.1 Multifaceted roles of pharmaceutical excipients.

Route of administration	Fundamental and popular excipients	Role and functionality	Reference
Oral	Cellulose and its derivatives, e.g., Avicel, microcrystalline cellulose	Assistance in rapid dispersion of tablets in the gastrointestinal tract promotes dissolution by increment in surface area.	Thoorens et al. (2014)
Tablets	Lactose, e.g., monohydrate lactose, super spray-dried lactose	Offers provision for adding bulk and facilitating accuracy in dose for potent APIs.	Molina et al. (2018)
	Starch and its derivatives, e.g., native starches, pregelatinized starch, sodium starch glycolate	Provides mechanical strength by binding ingredients of a tablet together.	Markl and Zeitler (2017)
	Polyols, e.g., mannitol, sorbitol, xylitol	Act as a sweetening agent and makes chewable tablets more palatable.	Riley et al. (2015)
	Silica and stearates, e.g., colloidal silica, granulated colloidal silicon dioxide, magnesium stearate	Imparts free flowability to powders during tablet manufacturing by reducing friction and adhesion between particles/granules.	Pandey et al. (2017)
Capsules	Microcrystalline cellulose (MCC), starch, and lactose	Commonly used for manufacturing of hard gelatin capsules	Thoorens et al. (2014)
	Natural vegetable oils, parabens	Primarily used for liquid-filled soft gelatin capsules	El Asbahani et al. (2015)
Solutions and sirups	Water, hydroalcoholic, polyhydric alcohols, buffers	Used as vehicles in solutions, sirups; categorized under aqueous vehicles.	Grembecka (2015)
	Vegetable oils, mineral oils, oily organic bases	Used as vehicles in solutions, sirups; categorized under oily vehicles.	El Asbahani et al. (2015)
	Glycerol or glycerin	Used as a sweetening agent in sirups	
Suspensions and emulsions	Hydrate carbons	Imparts easy swelling, enhances dissolution in water to form a colloid having high viscosity.	Ahmed et al. (2014)
	Gums, e.g., acacia, tragacanth	Acts a binder and fastens emulsification effect.	Hamman et al. (2015)
	Sterols, e.g., lanolin, wool fat	Used as emulsifiers in w/o emulsions	Weatherby and Carter (2013)
	Phospholipids	Used as emulsifiers in o/w emulsions	Van Hoogevest et al. (2014)

(*Continued*)

TABLE 19.1 (Continued)

Route of administration	Fundamental and popular excipients	Role and functionality	Reference
	Alginates, methylcellulose, carrageenan, gelatins, powdered cellulose, carbomers	Used as suspending agents in the preparation of suspensions.	Rabiskova et al. (2004)
Topical and transdermal	Hydrocarbon bases (oleaginous), e.g., white petrolatum, hard paraffin, soft yellow paraffin, and gelled oleaginous vehicle.	Act as occlusive and emollient.	Conway and Brown (2014)
	Absorption bases, e.g., cholesterol, lanolin, lanosterol, and hydrophilic petrolatum	Enhance the permeation of the drug upon the incorporation of a water-in-oil emulsion.	Jankowski et al. (2017)
	Water-soluble bases, e.g., polyethylene glycols (PEGs), polyoxyl 40 stearates, and polysorbates	Primarily used for incorporation of solid substances in topical or transdermal formulations.	D'souza and Shegokar (2016)
	Emulsion bases, e.g., lanolin, cold cream, and hydrophilic ointment	Used for emollient and occlusive effects and absorption of additional water.	
Parenteral	Water for injection *USP*	Used as the most popular vehicle for injectable preparations.	D'souza and Shegokar (2016)
	Vehicles for parenteral formulations, e.g., Ringer's injection and sodium chloride injection	Used as isotonic vehicles for the addition of active agents at the time of administration.	
	Water-miscible solvents for parenteral formulations, e.g., ethyl alcohol, propylene glycol, and polyethylene glycols	Used to enhance the solubility of certain active agents and for reduction of hydrolysis.	
	Nonaqueous vehicle for parenteral formulations, e.g., corn oil, cottonseed oil, peanut oil, and sesame oil	Generally, used for the parenteral hormonal preparations, e.g., testosterone injection.	Woon and Fisher (2016)
	Buffers, e.g., acetate, citrate, and phosphates	Used for stabilization of solutions against chemical degradation, especially in the conditions of significant pH alterations.	
	Antioxidants, e.g., sodium bisulfide	Used for the protection of injectable products against oxidative degradation.	
	Antimicrobial agents, e.g., benzyl alcohol	Generally, used to prevent the growth of bacteria and other organisms.	

(*Continued*)

TABLE 19.1 (Continued)

Route of administration	Fundamental and popular excipients	Role and functionality	Reference
Mucosal, vaginal, and rectal preparations	Bases in the preparation, e.g., cocoa butter, glycerinated gelatin, hydrogenated vegetable oils, mixtures of polyethylene glycols of various molecular weights, and fatty acids esters of polyethylene glycol	Used for the formation of the inert matrix of suppositories.	Fatmi et al. (2021)
	Glycerinated gelatin	Generally, used as a vehicle for vaginal suppositories.	

19.2.4 Quality aspects

The safety and effectiveness of any therapeutic formulations critically rely on the quality of excipients used. In this context, along with consideration in manufacturing variables to a suitable standard of GMP, it is also significant for manufacturers to assure the quality of excipients on the grounds of the specification, analytical measures, and sampling techniques (Saluja and Sekhon, 2016). It is essential for all parties engaged in supply chain management of excipients to be aware of the quality and grade. Quality management, supply chains, and various development schemes consist of numerous challenges associated with them. Not resolving these challenges creates opportunities for substandard quality excipients to enter into the supply chain.

Over the past few years, several cases of toxicity due to excipients have been reported in patients undergoing medication for various conditions; popular ones include propylene glycol (used as solvent intravenous (IV) administration of lorazepam and diazepam) toxicity characterized by hyperosmolarity and high anion gap metabolic acidosis, often accompanied by injury of kidneys and multiorgan failure (Tayar et al., 2002). In another reported incident from China, about 78 million medicinal gelatin capsules contained chromium, a carcinogenic heavy metal.

Similarly, the US FDA Center for Drug Evaluation and Research (CDER) banned certain phthalates as excipients in CDER-regulated products due to severe human health risks associated with exposure to dibutyl phthalate. Inappropriate regulation of excipients globally and poor supply chain management has eventually resulted in the potential malpractice of intentionally adulterated excipients, which ultimately turns into a life-threatening global issue for patients. Hence, critical supervision of excipient production and supply chain is vital to ensure high-standard excipients in the market. In this perspective, IPEC and its four regional centers, that is, IPEC-Americas, IPEC Europe, IPEC Japan, and IPEC China, are working as a bridge between excipient manufacturers to ensure the best quality of excipients.

19.2.4.1 GMP for excipients

IPEC-Americas and IPEC Europe, with other member companies, had launched the first voluntary standard for excipient GMP in the year 1995. Furthermore, in 2003, in collaboration with a Pharmaceutical Quality Group (PQG), they had agreed to produce novel and updated guidelines for manufacturing high-quality excipients of pharmaceutical use. These guidelines consist of every detail for GMP for bulk pharmaceutical excipients with PQG's PS 9100:2002 along with a memorandum of assurance to excipients manufacturers/consumers that produced excipients must be internationally accepted GMP quality. GMP standards for excipients address three fundamental difficulties, that is, safety, quality, and functionality that are more important than the efficacy of APIs. The progress of novel pharmaceutical ingredients is gradual due to the lack of global specific advice to analyze and ensure the safety of new excipients in pharmaceutical formulations.

In addition, the pharmaceutical industry required innovation in excipients that can increase the quality and efficacy of the formulation. The importance of excipient GMP compliance was raised by the US Food and Drug Administration (FDA) in 2008. Currently, there are no global regulation standards for excipients, which means that regulations are scattered, although drug authorities are working to improve the regulatory status of pharmaceutical excipients.

There are various kinds of regulatory bodies involve in the regulations of pharmaceutical excipients. They are given as follows:

- International Pharmaceutical Excipient Council Europe
- Japanese International Pharmaceutical Excipient Council
- International Pharmaceutical Excipient Council of America.

The World Health Organization (WHO) began globalizing pharmaceutical rules by focusing on rational medicine usage, medicine cost, and drug safety financial assistance for healthcare as well as an effective healthcare system. To deal with the impact of globalized medicine supply, globalized good manufacturing practice criteria for pharmaceutical excipients are critical systems of medicine distribution (Chua et al., 2010).

19.2.4.2 Good distribution practices for excipients

In 2006, IPEC-Americas and IPEC Europe published a guide entitled "Good Distribution Practice Guide for Pharmaceutical Excipients" for manufacturers engaged in producing pharmaceutical excipients. This guide works on a two-bay basis, that is, first, it renders additional, comprehensive information to the WHO technical report. Second, it is also aware that the manufacturers must provide a certificate of analysis (COA) to consumers. As COA assures that the consumers that produced excipient meet the significant standards of quality as per required specifications.

In addition to this, IPEC has also developed an excipient information package, which assimilates the information and relative source of data of excipient qualification with the standard package. The most significant advantage of this package is that it eliminates the massive load of surveys from multiple customers (Saluja and Sekhon, 2016). Good distribution practices are mainly part of quality assurance consisting of purchasing, storing, supplying, exporting, and importing pharmaceutical excipients in practice and not dealing with dispensing of these excipients or products directly into market or patients. The parties that are busy in the distribution of pharmaceutical excipients should confirm the quality of pharmaceutical excipients and the process of distribution maintained with integrity from manufacturer to agent or party. National legislation and guidelines are applicable for good distribution of the pharmaceutical product in the region or country as possible, to put and establish minimum standards.

19.2.4.3 Excipient biological evaluation guidelines

The excipient biological evaluation guidelines are generally for determining the safety of APIs and excipients. APIs and excipients differ from each other due to their specific functions and mechanism of action. Hence, the biological activity of APIs and excipients also vary from each other. When any novel excipient enters into the pharmaceutical market, identifying their safety is mandatory to get approval from intended regulatory market. In all aspects, the excipient's safety test procedures should always adhere to the Guiding Principles on the Use of Animals in Toxicology and Good Laboratory Practice. The safety profile of various pharmaceutical excipients was assessed through the analysis of the database available and the literature. To develop a database with certain information for each excipient used in the pharmaceutical business, all relevant information, both chemical and toxicological, must be collected, including the probable routes of administration, dose, and exposure time. In a simple term, the physical and chemical attributes of the material, the production procedures that can be employed, the routes of administration, the dosage form, and the target population must all be included in the safety information. The limit range of toxicity, impurities, drug−excipient interaction, and exposure conditions should be investigated (Kim et al., 2016).

In addition, pharmacokinetic and ADME studies are the critical factors for the choice of excipient. Moreover, the age group like pediatric and geriatric needs more focused on manufacturing pharmaceutical formulations that can be controlled by their requirement and capacity to deliver the drug (suspension and solution). Moreover, in these patients, the excipients may give rise to adverse or toxic effects than another type of population since variation in biotransformation and elimination could result in harmful consequences.

The microbiological growth in suspension or solution can be controlled by applying temperature throughout the formulation process. In addition, the temperature is also used for identifying and confirming the dissolution process in the product. In oral suspension, the additional identification includes particle size of drug in suspension, the dissolution, the pH, and viscosity related to bioequivalence study (Zupančič et al., 2016). The excipient biological evaluation guidelines give the preliminary information to produce an authentic database for new excipients.

19.3 Need for excipient compatibility studies between excipients and drugs

A drug−excipient compatibility study signifies a crucial phase in the formulation and development of the pharmaceutical product. Certain criteria/properties such as chemical structure, purity, authenticity, type of delivery system required, and the projected manufacturing methodology need to be assured before converting a drug substance into a dosage form. Proper and thorough compatibility studies must be conducted since the risk or chances of chemical reactions or interactions between drugs and excipients cannot be neglected. In this case, compatibility studies act as a promising tool in deciding whether to opt for an excipient for the formulation or not. This permits the speedy optimization of a dosage form concerning patentability, dispensation, drug release, elegance, and physicochemical stability. To acquire the rapid stability profile of drug and excipients, stability studies are examined under the stress conditions as per standardized procedures and/or preestablished information on potential degradation pathways or incompatibilities (Dinte et al., 2013). The ideal properties of excipients are presented in Fig. 19.1.

19.3.1 Compatibility studies between excipients and drugs

Drug–excipient compatibility studies are essential because of the following rationales:

- It assists in maximizing the stability of a dosage form, as any physical or chemical interaction between drug and excipient can affect the bioavailability, stability, and safety of a drug.
- It supports in avoidance of unanticipated concerns.
- It assists in forecasting the probable reactions between drugs and excipients, which saves the formulated dosage forms from detrimental effects.
- By utilizing compatibility studies, one can opt for the optimum excipient type with the chemical entities promising in drug discovery programs.
- A drug–excipient compatibility study is vital for investigational new drug submission since USFDA has launched new regulations regarding the approval of new formulation systems that all submissions must consist of drug–excipient compatibility studies.

Some critical issues associated with the postformulation system can be prevented; for example, sucrose/lactose should be used in drug substances of primary amine nature

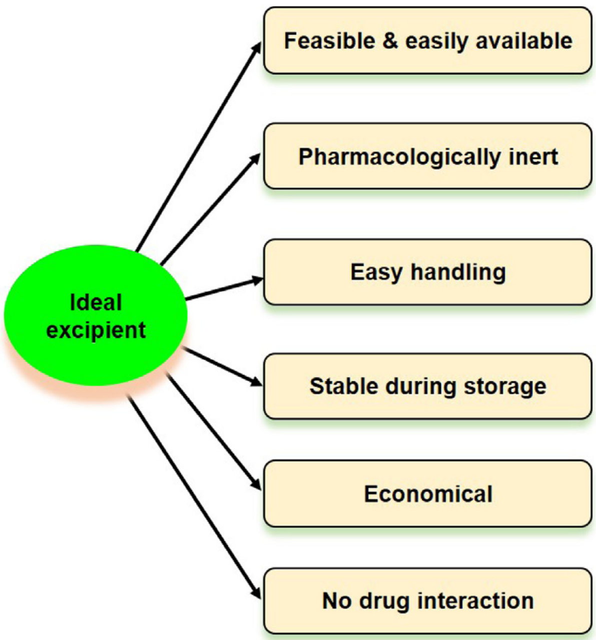

FIGURE 19.1 Ideal properties of excipients.

since the system has the probabilities of interactions that could form a colored compound. The computational programs such as SPARTAN, CAMEO, PharmD3® EPWIN, and other internal databases help to determine the degradation path of the compounds. The compatibility studies between drug and excipient usually involve a physical mixture of drug and excipients. The quantity and number of excipients in the physical mixture should be high compared to the formulation for evaluating reacting species in excipients responsible for producing incompatibility in formulation (Chadha, Bhandari, 2014). Polypharmacy is associated with the administration of several drugs at a time that results in possible pharmacokinetic interaction between drugs as well as excipients in the formulation. Sugar alcohols such as sorbitol and mannitol are used as excipients in pediatric formulations responsible for lowering the oral bioavailability of several drugs (Matsui et al., 2021).

19.3.2 Stability determination of excipients

The stability determination of excipients mainly depends on the category of excipients used and their physical and chemical attributes. If more than one excipient is used in the formulation development, it is essential to understanding the drug–excipient interaction and excipient–excipient interaction (Jahangiri et al., 2016). According to the stability conditions, the excipients were categorized as stable, very stable, or having limited stability. These categories of excipients are based on the period in which excipients are stable when closed in a particular container. The stability determination consists of both chemical and physical attributes determination of excipients. When excipients are categorized as very stable, they remain unaffected throughout the production process and possess the same quality for at least 5 years in a specific package. On the other hand, stable excipients exhibit stability between the periods of 2–5 years.

Moreover, some excipients showed limited stability because of their reassessment interval and expiration date of fewer than 2 years. In addition, these kinds of excipients are highly vulnerable to moisture absorption, hydrolysis, change in viscosity, oxidation, degradation by light, or due to affected by various environmental conditions. The excipients that absorb moisture, loss on drying, or assessing moisture content can be used as criteria for determining stability testing. It is essential to confirm the stability of the excipient during the manufacturing process since assurance of excipient protection in the packaging process and storage is applicable to confirm excipient protection, to circumvent the degradation compound formation resulting toxicity. Interestingly, individual excipients can be categorized in one or more classes of stability based on packing, which must be uniform and always justified (Abrantes et al., 2016).

19.3.3 Impurities in pharmaceutical excipients

Reactive impurities in pharmaceutical excipients may result in product degradation and affect the quality and stability of drug products. A most common example is tween 80, which is widely used as a surfactant in various formulation development that contains many impurities like formaldehyde, fatty acids, peroxides metals, etc., untoward effects of

TABLE 19.2 Reactive impurities in pharmaceutical excipients.

Reactive impurities in pharmaceutical excipients	Examples of excipients containing reactive impurities	Type of chemical reaction	Incompatibility in excipients	Reference
Peroxides	Organoperoxides, hydrogen peroxide, hydroperoxides	Oxidation of drug substance and starts polymerization reaction	Raloxifene oxidation in the presence of povidone, which contains peroxides.	Dave et al. (2015)
Reducing sugars	Sucrose, glucose, mannitol, and maltitol	Condensation reaction, Maillard reaction, Amadori rearrangement reaction	Impurities present in glucose from microcrystalline cellulose (MCC) results in discoloration because of the interaction of L-phenylalanine of drug substance complex	Wu et al. (2009)
Aldehydes	Povidone, hypromellose starch PEG, polysorbate, and HPC	Generally cross-linking with amino-functional group	Formaldehyde forms complex with amino groups in hydroxymethyl species, and hence formaldehyde in tween 80, PEG 300 enhance the degradation in protein-based formulations	Toews et al. (2010)
Organic acids	Polyvinyl alcohol, PEG	Reaction of alcohol with the formation of esters takes place	Formate species are found in polymers such as PVA, and PEG results in the formation of ROS and free radicals.	Hemenway et al. (2012)
Elemental impurities	Copper and iron	Metal catalyzed reaction	Iron is responsible for oxidation and deamidation reactions	Ji et al. (2009)
Other reactive impurities (nitrate and nitrite)	Starch, microcrystalline cellulose (MCC), sodium starch glycolate, povidone lactose, crospovidone	Formation of *N*-nitroso compounds and deteriorate the drug substance	Bromhexine results in nitrosation reaction.	Zhang et al. (2018)

these impurities lead to particulate formation or protein oxidation. The various types of reactive impurities in excipients are discussed in Table 19.2 (Gorain et al., 2018).

19.4 Mechanisms of excipient and drug interactions

In almost all the pharmaceutical dosage forms, excipients have a larger quantity than APIs. For example, tablets, a considerable number of excipients, are evidenced by various categories of excipients such as a binder, disintegrant, lubricant, and diluents. Therefore excipients may have a tremendous impact on the performance of API. For example, in solid dosage forms, excipients directly impact the safety and efficiency of API either by promotion or impediment in a gastrointestinal release. Hence, the study of interactions between excipients and drugs is critical before the selection of formulation of a dosage form (Chowdhury et al., 2018).

APIs have varying structural attributes that correlate with receptors or assist in metabolic activities. These unavoidably present risks that make them susceptible to various degradation modes; some of the most common methods are discussed below.

19.4.1 Hydrolysis

Medicinal agents having functional groups such as amides, esters, and lactones are vulnerable to hydrolytic degradation. Hydrolysis is one of the most common modes of degradation among API. The presence of water also facilitates microbial growth (Alsante et al., 2014). Hydrolysis is a type of reaction that consists reaction of active ingredient and water with broad pH range. This type of reaction ends with the formation of a new product that affects the stability of the active substance. Hirakura et al. found that simultaneous ester formation due to lactic acid and propylene glycol or ethanol and hydrolysis of a dimer of lactic acid enhanced the pH of parenteral formulation (Hirakura et al., 2006).

19.4.2 Oxidation

In contrast to hydrolysis, oxidative degradation is more intricate, which engrosses removal of an electropositive atom, radical, or electron or, on the contrary, the inclusion of an electronegative moiety. Oxidation reactions are catalyzed by oxygen and heavy metal ions, which lead to the formation of free radicals. Furthermore, these free radicals interact with oxygen to form peroxy radicals, which react with oxidizable compounds to liberate extra free radicals (Lobo et al., 2010). The various functional groups such as alcohols, aldehyde, phenols, unsaturated fatty acids, and alkaloids are highly vulnerable to the oxidation of compounds. Kollmer et al. demonstrated that the degradation of benzocaine follows the oxidative degradation pathway because of the presence of reactive impurities in tablets such as mannitol, crospovidone, and microcrystalline cellulose (MCC) (Köllmer et al., 2013).

19.4.3 Isomerization

The conversion of a chemical into its optical or geometric isomer is isomerization. This transformation produces those isomers that have different pharmacologic or toxicologic properties. For example, the performance of the levo (L) form of adrenaline is far better than its dextro (D) form (Nguyen et al., 2006). Fathima et al. reported that when adrenaline, vitamin A, and tetracycline undergo isomerization process results in the formation of new isomeric forms that has affected the therapeutic activity of compounds (Fathima et al., 2011).

19.4.4 Photolysis

Whether natural sunlight or artificial light, it directly accelerates the catalysis of various reactions mentioned earlier. Most drugs absorb ultraviolet light, so degradation due to lower wavelength is quite common. Exposure to light leads to discoloration even when chemical alterations are imperceptible. In contrast to excipient–excipient interaction, drug–excipient interactions occur more frequently, which may be advantageous or unfavorable. Drug–excipient interactions are also categorized on a physical and chemical basis. Hubicka et al. (2013) investigated the photodegradation of fluoroquinolones derivatives such as norfloxacin, ciprofloxacin, ofloxacin, and moxifloxacin by UPLC/MS-MS analysis. It was seen that photodegradation occurs due to the compounds such as TiO_2 and Fe_2O_3 (Hubicka et al., 2013). Some of the compounds that flow photolysis include nifedipine, folic acid, and riboflavin, etc. Interestingly, if an excipient with antioxidant action consisting of metallic impurities potentiates the degradation of the active ingredient.

19.4.5 Chemical interactions

These interactions involve a chemical reaction between drugs and excipients or impurities present in them. Chemical interactions are usually damaging to the dosage form since the degradative products get liberated after the chemical interaction. Various degradation product is categorized in ICH guidelines ICHQ3B (Teasdale et al., 2017). In the presence of an aqueous environment, effervescent interactions between sodium bicarbonate and citric acid produce carbon dioxide (CO_2), which aids in the disintegration of the solid dosage form for oral intake of the integrated medicinal component. The precaution should be taken while storing any pharmaceutical product to circumvent the moisture contact with the product to prevent chemical interactions mediated by the water. According to International Conference on Harmonization (ICH) guidelines, monitoring can be done on products that include limits of various impurities and their evaluation in marketed medicinal agents (Gorain et al., 2018).

19.4.6 Physical interactions

These interactions are quite challenging to detect and are occurred unintentionally, which may be beneficial or detrimental. One of the famous examples of physical interaction is between primary amine drugs and MCC. It has been found that drug release gets slower or not released from the dosage form due to binding with MCC (Fraser Steele et al., 2003). The primary purpose of carrying out physical interaction is to increase the solubility of the active ingredient. Since untoward interaction affects the activity of the compound, influence on dissolution rate may directly impact the drug product's bioavailability and thus its efficacy, which, in turn, impacts the patient who is taking the drug. The beneficial impacts of excipients' functional properties are also examined at the same time to make it easier to make parenteral formulations, ophthalmic therapies, and/or open wound treatments for a given medicine. In addition to this, the microbiological content, impurities, and bacterial endotoxin investigation are most important. Drug complexation with cyclodextrin or its derivatives is a common example of establishing physical interaction to improve oral bioavailability.

Another example of interactive physical mixing is the addition of very fine magnesium stearate particles to the tablet or capsule formulation. Many formulation scientists have proved the coating of magnesium stearate particles, which creates an aqueous environment barrier due to its water-repellent qualities (Gorain et al., 2018).

19.4.7 Role of residual solvents and impurities in interactions

Excipients came from a diversified array of origin, which further gets processed with the help of various chemical/physical treatments to acquire a final finished product. Due to this processing, residues are perpetually left after isolation. For example, dextrose is extensively utilized as a tonicity enhancer in the parenteral dosage forms, and it is used as a nutrition solution. During sterilization exercise of the dosage form, dextrose usually undergoes isomerization. It gets transformed into fructose, which further reacts with the primary amino group to form a Schiff base and causes discoloration (Zhang et al., 2018). The residues may be present after isolation of the compounds, and a very small quantity of residues dramatically affects the product safety and activity. We discuss various residues in different excipients in Table 19.3.

19.4.8 Interaction's evaluation in drug and excipient

The chemical and physical interactions are most commonly observed in drugs and excipients than in excipients. The chemical interaction is mainly responsible for the deterioration of drug substances or the formation of new impurities in the formulation. While the physical interaction may result in a change in the rate of dissolution or dosage form uniformity and bioavailability of prepared formulation. Hydrolysis is the principal reaction that occurs in liquid-based formulations. Other than hydrolysis, the drug substance is also prone to photolysis, isomerization, and polymerization reactions. These kinds of reactions reduce the potential of APIs and may form impurities in the formulation.

19.4.8.1 Evaluation by computational method

The interaction between drugs and excipients can be easily evaluated by the computational method that uses an available database for interaction evaluation. The database available mainly contains comprehensive information regarding the excipient, functional group present in an excipient, and impurities present. The excipient safety profile test that can be done by quantifying method is very rapid, and no large amount of sample is required. The limitation of this method is very low predictability (Bharate et al., 2016).

TABLE 19.3 Residues and impurities found in general excipients.

Excipients	Function	Examples of preparation containing this excipient	Impurity or residue
Calcium phosphate dihydrate (dibasic)	Diluent and binder	Tablet	Alkaline matter
Benzyl alcohol	Preservative and solubilizing agent	Solutions and injectables	Benzaldehyde
Magnesium stearate	Lubricant	Tablet and capsule	Antioxidants
Starch	Binder and disintegrant	Tablet and solid dosage forms	Formaldehyde
PEG (polyethylene glycol)	Solvent, surfactant, and plasticizer	Ointment, suppository, tablet and capsule	Peroxides and organic acids
Stearate derivatives	Lubricating agent	Tablet and capsules	Alkaline matter
Povidone, crospovidone	Binder and disintegrant	Capsule, tablet	Peroxides
Cellulose derivatives	Binder	Tablet	Glyoxal
Lactose	Filler or diluent	Tablet and dry powder inhalations	Reducing sugars
Fixed oils	Stabilizer	Parenteral preparations	Antioxidants
Talc	Glidant and lubricant	Tablet and cosmetic preparation	Heavy metals
Lipids	To solubilize hydrophobic APIs	Tablets and solid dosage forms	Peroxides
MCC (microcrystalline cellulose)	Disintegrant	Tablet	Hemicellulose, lignin
Polysorbates	Surface active agent and solubilizer	Cosmetics and ophthalmic preparation	Peroxides

19.4.8.2 Isothermal microcalorimetry

Isothermal microcalorimetry is the technique used to identify changes in the heat flow. Hence, it is easy to detect any changes in the mixture of API and excipients, affecting the stability of the formulation. When the change was observed by isothermal microcalorimetry technique, Scanning Electron Microscopy and Hot Stage Microscopy were performed to explain the problem. High-resolution mass spectrometry, HPLC, or liquid chromatography can obtain more precise and promising results (Abrantes et al., 2016).

19.4.9 Analytical techniques used to characterize drug/excipient compatibility

The characterization of drug−excipient interaction is critical in formulation development to accomplish bioavailability, stability, and manufacturing capacity of solid dosage formulations. Most commonly used analytical tools used for compatibility studies include thermogravimetric analysis, differential scanning calorimetry, hot stage microscopy, isothermal microcalorimetry, powder X-ray diffraction, scanning electron microscopy,

TABLE 19.4 Tools for compatibility screening of drugs with their advantages and limitations.

Tools	Merits	Limitations	Reference
Differential thermal analysis (DTA) and differential scanning calorimetry	Requires lesser amount of sampleFaster methodIt detects the interactions by determining the polymorphic nature of the substance.Identifies crystalline or amorphous nature of compounds. Eutectic formation can be detected by the DTA technique.	It involves sample degradation, i.e., destructive method.Not applicable for compounds in which thermal differences are negligible.Unable to identify incompatibilities that appeared after a long-term period.	Mura et al. (1998)
Isothermal stress testing	Helps to determine identification and evaluation of impurity throughout the incompatibility.	Tedious and more time required for the completion of the process. For HPLC analysis, it requires method development for individual sample.	Phipps et al. (1998)
Powder X-ray diffraction	Identifies the incompatibility by the presence of specific diffraction peaks. Physical incompatibilities detected by changes in a crystalline or amorphous state. Method can be used qualitatively as well as quantitatively. The method is nondestructive.	Unable to explain the chemical structure. The diffraction pattern has preferred orientation.	Harding et al. (2008)
Scanning electron microscopy	Minimal sample size required greater resolution as compared to light microscopy to investigate microscopical images.	Sample preparation required. Requires vacuum setting.	Chadha and Bhandari (2014)
HPLC and TLC	The degradation of compounds is well defined. The spots that belong to the degradation of compounds can be forwarded for possible detection.		Chadha and Bhandari (2014)
Hot stage microscopy	Physical changes may be observed when drug contact with an excipient.	Sample degradation occurs as the method is destructive. Sample preparation needed	Chadha and Bhandari (2014)
Solid-state NMR nuclear magnetic resonance spectroscopy	Expensive Large sample required Time-consuming		Chadha and Bhandari (2014)
Diffuse reflectance spectroscopy	The decomposed products can be detected.		Chadha and Bhandari (2014)

Fourier-transform infrared spectroscopy, and high-performance liquid chromatography, and novel tools include NMR and NIR that are having broader applications in characterization of solid substance (Table 19.4). Hence, these kinds of techniques are widely used in the evaluation of drug−excipient compatibility.

19.5 Concern on off-label formulation excipients

The "off-label" is the term that indicates that the drug products used are different than that explained in the license. As mentioned earlier, there is no specific type of medication available for pediatric as well as geriatric patients, so using adult medicines leads to significant adverse reactions and toxicity. The examples include in off-label formulation, such as administering medicine with more significant dose than mentioned in license and taking medicine by pediatric and geriatric patients where a licensed range of dose is not available. These kinds of conditions may worsen the health of patients if the adverse effects of therapeutic agents are more potent. Special care has to be taken while taking "off-label" formulation, and advice from doctors and experts is essential to prevent any toxic reactions.

19.5.1 Preservatives

Liquid oral dosage forms are the primary alternatives for children among the different formulations. The advantage associated with liquid dosage forms includes dose flexibility, comparatively easy swallowing. However, liquid formulations demand the usage of excipients that are not good for children's health, such as preservatives (Binson et al., 2019). It is unavoidable to prevent microorganism growth without adding preservatives in the formulation (Dresser and Frader, 2009). Preservatives are the agents added in formulation to prevent the growth of microorganisms and assure long-term storage of pharmaceutical products. The most common example includes sucrose that acts as a self-preservative in sirup formulation and inhibits growth of microorganism.

19.5.1.1 Parabens and their derivatives

Parabens are the most widely used preservatives in pharmaceutical formulations. However, derivatives of parabens such as methyl, ethyl, and propyl parabens used in pediatric formulations may be risky as they affect the bilirubin-albumin binding. These preservatives can be fatal in neonates suffering from hyperbilirubinemia. Paraben-containing formulations are contraindicated in infants who have jaundice. Andersen et al. revealed that butyl parabens show cell damage and skin sensitization in animals (Andersen, 2008). By considering the taxological effects of parabens and their derivatives, WHO suggested a total acceptable daily limit of not more than 10 mg/kg/day (Yang and Purchase, 1985).

Moreover, parabens and their derivatives are also used in personnel care products as preservatives. Cosmetic products like lotions, shampoos, deodorants, and scrubs contain parabens as a preservative. The parabens are an odorless and white crystalline substance with a slightly metallic taste. These are resistant to hydrolysis and can work well in the pH range of 4.5−7.5. However, the mechanism of action of parabens is nonspecific. As mentioned earlier, parabens are widely used in cosmetic-based products, and many people contact with parabens. One case study in the United States found that parabens were observed in almost all urine samples of US adults despite socioeconomic and ethnic background. In addition to this, parabens are also responsible for producing an allergic reaction. However, the effect varies from individual to individual. In recent studies, it has been observed that parabens resemble estrogen in structure, show weaker estrogenic activity, and also interfere with hormonal changes in the male fetus and young children (Garner et al., 2014).

19.5.1.2 Benzyl alcohol

Because of its local anesthetic property, benzyl alcohol is used in the solutions and injectables majorly. It is suspected that benzyl alcohol used in intravascular and endothelial tube lavage formulations leads to the deaths of underweight, premature infants, and infants suffering from metabolic and respiratory disorders. The use of benzyl alcohol and benzoic acid in IV intramuscular (IM) formulations causes metabolic acidosis and hemorrhage, and thereby increases the mortality rates in the neonates (Burgess, 2018). By switching these preservatives with the alternative preservatives significantly prevented acidosis and hemorrhage in neonates. Of note, 130 mg/kg or more significant exposure to benzyl alcohol in neonates leads to the gasping syndrome, characterized by hypotension, metabolic acidosis, bradycardia, and metabolic acidosis.

A case report investigation revealed that IV administration of benzyl alcohol at a dose of 99−234 mg/kg/day in the 1-year-old neonates via large volume parenteral leads to death. The benzyl alcohol adverse effects such as convulsions, respiratory distress, paralysis, and vasodilation have been known for years. However, in premature infants, the toxic effect of metabolic products of benzyl alcohol is not known clearly.

In another study, when 32−105 mg/kg/day dose of the benzoic acid leads to neonatal death due to respiratory and metabolic complications upon IM injection of benzoic acid. WHO recommended a 5 mg/kg body weight dose for the benzyl alcohol and benzoic acid as an acceptable limit. However, the dose of these preservatives in pediatric patients is not yet well established.

19.5.1.3 Benzalkonium chloride

It is majorly used in ophthalmic formulations. Around 74% of the ophthalmic formulations contain benzalkonium chloride as a preservative. It is also used in antiasthmatic drugs and nebulizer formulations as a preservative. However, the use of benzalkonium chloride causes bronchodilation in the pediatric population. Sometimes this leads to the precipitation of respiratory arrest. Prolonged use of benzalkonium chloride in topical formulations and applied on the ear may cause ototoxicity and hypersensitivity. Benzalkonium chloride is bactericidal in action and related to problematic spasm of bronchi in asthmatic patients after administration of medication. Benzalkonium chloride is most commonly found in ipratropium bromide and beclomethasone containing nebulizer formulations and is responsible for long-term bronchoconstriction by non-IgE mechanism (Fabiano et al., 2011).

19.5.1.4 Sodium benzoate

Sodium benzoate is widely used in various preparations such as vitamin preparations, antibiotic sirups, and anticough sirups. It is implicated at the beginning of the various types of food-induced urticaria, asthma, and anaphylaxis. In recent studies, it has been observed that sodium benzoate produces hypersensitivity reaction in children following cutaneous type of reactions that occurs when the administration of suspension containing amoxicillin with clavulanic acid occurs. Some reports suggest that benzoates are similar in structure to acetylsalicylic acid, and administration of benzoates may result in the production of eicosanoids (Barbaud, 2014).

19.5.2 Solvents

Although water is considered the universal solvent for preparing liquid formulations, most of the APIs suffer from low water solubility, which restricts the achievable concentration in the formulations. In many instances, cosolvents and surfactants like ethanol, polyethylene glycol, and glycerol are required to enhance the drug solubility and thereby achieve the required drug concentration in the solution formulations. However, the use of these excipients requires safety consideration as these excipients significantly influence pediatric health. Solvents and solubility-enhancing agents generally cause tissue damage, irritation, and local site toxicity. The risk is very high when these are included in the parenteral formulations instead of oral formulations (Batchelor and Marriott, 2015). Solvents are mainly used in the formulation of liquid dosage forms.

In addition, cosolvents can be incorporated in liquid-based preparation that acts as a vehicle for APIs. Many types of excipients are utilized in the preparation of parenteral formulation. Some of the solvents are listed in Table 19.5.

19.5.2.1 Propylene glycol and polyethylene glycol

Polyethylene glycol is generally used as a preservative, cosolvent, and humectant (Mason et al., 2012). These polymers are degraded by the aldehyde dehydrogenase enzyme. However, in infants, these enzymes will not present sufficient to eliminate the PEG, leading to side effects such as hemolysis and hypotension. Kaplan et al. investigation showed that applying the dexamethasone cream containing 8% (w/w) PEG on the wounded surface results in hyperosmolarity, which leads to a respiratory arrest (Kaplan et al., 2013). Abound 5.2% of the pediatric patients were affected with hyperosmolarity when they took topical formulation (Patra et al., 2016).

Kristine et al. revealed increased gastric emptying time of the ranitidine and cimetidine formulations containing PEG in children less than 3 years, thereby significantly reducing the bioavailability of both the drugs (Valeur et al., 2018). The results from this investigation demonstrate the reduced bioavailability of ranitidine and cimetidine due to the influence of PEG (Basit et al., 2002). In addition to this, PEG can also influence the metabolism of simultaneously administered drugs. One study found that PEG stearate induced the Cytochrome P450 3A4 enzyme and thereby reduced the total area under the curve of the midazolam (Ren et al., 2009).

TABLE 19.5 Solvents used in parenteral preparation.

Solvent	Scale
Peanut oil	Variable
Water	Variable
N,*N*-dimethylacetamide	6%–33%
Ethanol	0.6%–100%
Alcohol	Variable
Sesame oil	Variable
Polyethylene glycol	Variable
Cottonseed oil	73.6%–87.4%
Glycerin	1.78%–32.5%
Benzyl benzoate	20%–44.7%
Safflower oil	5%–10%
Dimethyl acetamide	1.89%–33%
Vegetable oil	Variable
Castor oil	0.0009%–0.03%
Soybean oil	10%
Benzyl benzoate	0.01%–46.0%
Propylene glycol	0.0025%–80%

19.5.2.2 Glycerol

Its palatability and suspending ability make glycerol one of the critical solvents in the suspension formulation. However, using more than 40 glycerol in the formulation contributes to side effects like headaches, mucositis, electrolyte disturbance, stomach upset, and diarrhea (Chatterjee and Alvi, 2014). These side effects of the glycerol may associate with the osmotic properties of the glycerol. Diarrhea caused by the glycerol may reduce the transit time and thereby affect the drug absorption in pediatrics (Yang and Purchase, 1985). Glycerol is generally considered as waste produced by the biodiesel industry. It is a green solvent that can be used in organic synthesis, catalysis, material chemistry, and separations. Glycerol can be readily available, nontoxic, and very low cost. It is a clear, odorless, colorless, sweet, and viscous liquid. It is a nonflammable and biodegradable solvent, and no specific treatment is needed while handling the solvent.

19.5.3 Antioxidants

Oxidation reaction in the dosage forms is one of the causes of drug instability. To prevent the oxidation reaction, many times, antioxidants are usually added into the formulations (Yochana et al., 2012). Propyl gallate and sulfites are widely available as off-label pediatric formulations. Antioxidants came into the picture by enhancing the shelf life of the pharmaceutical product. In a simple term, oxidation involves the addition of oxygen in the structure of the drug, which changes the structure of a compound having a lower number of electrons. The widely acceptable antioxidants include butylated hydroxytoluene, butylated hydroxyanisole, cysteine, propyl gallate, and sulfites. However, they are associated with several toxic effects such as gastric disturbances, diarrhea, and tumors. Antioxidants can be evaluated by various methods such as oxygen radical absorbance capacity assay, trolox equivalent antioxidant assay, and 2,2-diphenyl-1-picrylhydrazyl assay (Celestino et al., 2012).

19.5.3.1 Propyl gallate

Propyl gallate is available in both crystalline and amorphous forms. It is a widely used antioxidant in pharmaceutical formulations like emulsions, fat, and oil (Al-Khattawi and Mohammed, 2014). In vitamin A formulation, propyl gallate is commonly used as an antioxidant. Propyl gallate causes side effects like erythema and pruritis. It was reported that this excipient causes cancer and hyperactive reactions if not used to certain limits (EMEA, 2006). It is widely employed in food, cosmetics, and pharmaceuticals for preparing various formulations. It was prepared by the chemical synthesis process.

19.5.3.2 Sulfites

Sulfites are frequently used antioxidants in pharmaceutical and cosmetic products. Although their use is common, these antioxidants pose clinical side effects in children. The side effects include diarrhea, flushing, abdominal pain, and dermatitis. It was reported that about 3%–10% is the prevalence rate for sulfite sensitivity. The severity of the hypersensitive reactions depends on the children's health condition. The mechanism behind the toxic effects of the sulfides is remained unclear (Vally and Misso, 2012).

Sulfide derivatives like sodium sulfite, sodium bisulfate, sodium metabisulfite, sulfur dioxide, potassium metabisulfite, and potassium bisulfite are the FDA-approved generally recognized as safe excipients for the use of pharmaceutical and cosmetic products (Chaudhari and Patil, 2012). However, these antioxidants suffer from different toxicities in the pediatric population. Compared to the other sulfites, metabisulfite hypersensitivity is the primary concern in children (Yang and Purchase, 1985). Sulfites are preservatives that have more comprehensive applications in the food and soft drink industry. However, sulfites administration is associated with several symptoms such as gastric symptoms, coughing, rashes, urticaria, wheezing, and several anaphylactic reactions. A case study of sulfite was done on asthmatic children, and reactions produced by sulfites were investigated. Interestingly, it has been found that sulfites are responsible for provoking asthma attacks in less than 10% of susceptible children that are highly sensitive to the sulfites (Steinman and Weinberg, 1986).

19.5.4 Sweeteners

Sweeteners are essential to maintain the taste and enhance the palatability of the oral formulation for pediatric patients. The amount of the sweeteners used as flavoring agents depends on the API properties. The justification for the use of sweeteners in the pediatric formulation must be clearly described for the product acceptability by the FDA for marketing. As the sweeteners are associated with the concerns in diabetes and renal insufficiency patients, the risk-to-benefit ratio must be assessed in those patients (Tandel, 2011). The different types of sweeteners available in the market enhance the palatability of pharmaceutical products and mask the unpleasant taste of the active ingredients. Examples of sweeteners include Aspartame, Dextrose, Erythritol, Fructose, Galactose, Maltitol, Mannitol, Sorbitol, Sucrose, Saccharin sodium, etc. (Melgardt de Villiers, 2009).

19.5.4.1 Sucrose

Sucrose is most widely used globally. It will absorb in the form of fructose and glucose in the small intestine upon hydrolytic reaction. Due to their toxicity concerns, sucrose is not recommended in children's and adults' pharmaceutical formulations. Recently, it has been found that sucrose can have the ability to reduce the dental plaque pH and thereby dissolves the enamel in the aged population (Bukhamseen and Novotny, 2014). This decreased pH may cause the degradation of the acid-sensitive drugs and thereby results in therapeutic failure.

In the pediatric population, natural sweeteners were found to be carcinogenic. If at all required to use in the pharmaceutical formulation, then strong justification is required for their use. The sucrose can alter the osmotic properties of the GI fluids, which leads to reduced bioavailability of the few drugs (Adkison et al., 2018). In patients suffering from fructose intolerance and diabetes, particular emphasis must be given to sweeteners in medicinal preparations used for pediatric formulations (Helin-Tanninen, 2010). USFDA suggests not to use sucrose in pediatric formulations (Passos et al., 2010).

19.5.4.2 Aspartame

It is a synthetic sweetener and is widely used in pediatric and adult pharmaceutical formulations. It is the combination of aspartic acid and phenylalanine. These amino acids are primarily seen in foods. It is odorless and white powder slightly soluble in water. However, it is susceptible to heat. As mentioned previously, it contains phenylalanine, and hence it should not be given to the patient suffering from phenylketonuria. It has synergistic action when used with other sweetening agents. Phenylalanine is harmful to a specific group of the pediatric population. Hence, the phenylalanine content on the label of some pediatric formulations is mandatory. The children who are suffering from the epileptogenic disorder and phenylketonuria use of phenylalanine cause adverse reactions. Another problem with aspartame is that it will react with sulfonamides when used together. It is recommended not to use more than 40 mg/kg/day for all age groups (Melgardt de Villiers, 2009; Yang and Purchase, 1985).

19.5.5 Fillers

Diluents are generally considered chemically inert and are primarily used in solid dosage forms to make up the bulk (Qi et al., 2009). Some reports revealed the adverse effects of diluents in the pediatric population upon long-term use of the medicine. The excipients like lactose and mannitol are used widely in the off-labeled pediatric formulations and are reported for their toxicity profile.

The incorporation of fillers in the formulation has particular merits, which are as follows:

- It enhances the flow properties of materials.
- Greater linkage.
- Direct compression can be easily possible.
- Adjustment in weight can be made.

Fillers can be classified based on their nature, such as organic filler, inorganic filler, and coprocessed fillers. As the name indicates, fillers are generally responsible for increasing the bulk of pharmaceutical formulation (Manchanda et al., 2018).

19.5.5.1 Lactose

Lactose was used as a filler and binder in pharmaceutical formulations and is present around 20% of the solid dosage forms in the market. The disadvantage of lactose is that it causes hypersensitive reactions when used in infants and children by pediatric formulations (EFSA Panel on Dietetic Products and Journal, 2010). Since the immature intestine has a reduced level of lactase, it cannot hydrolyze the lactose into its hydrolytic products. This leads to lactic acid deposition and causes abdominal pain, diarrhea, and eczema (Balbani et al., 2006). The deposition of the lactose may sometimes change the osmotic pressure at the colon site, thereby reducing the absorbance of nutrients and drug molecules from absorption. WHO suggested not ingest greater than 3 g/day for children. Lactose can be categorized as follows:

- spray-dried lactose,
- lactose monohydrate (hydrous), and
- lactose anhydrous.

Spray-dried lactose is a free-flowing, and direct compression can be possible. However, it has a high cost when compared with hydrous lactose and anhydrous lactose. On the other hand, anhydrous lactose can be compressed directly, but it has inferior flow attributes. Moreover, lactose monohydrate can be employed in the wet granulation process and economic (Manchanda et al., 2018).

19.5.5.2 Mannitol

Mannitol is generally used in chewable tablets and orodispersible tablets. Relatively it is less carcinogenic compared to the sugars. Mannitol is costlier and hydrophilic, has poor flow property, and is noncarcinogenic. The adverse reactions like anaphylactic shock and hypersensitivity were observed after infusing the 10% or 20% (w/v) mannitol intravenously (Nagpal et al., 2016). Anaphylactic reactions are observed due to the direct action of the mannitol on mast cells. Oral absorption of mannitol is not more than 20%. Since mannitol can alter osmotic pressure, it can influence the intestinal transit time, thereby altering the bioavailability of the drugs (Narang, 2015).

19.5.6 Miscellaneous agent

19.5.6.1 Dyes

The appropriateness of hypersensitivity reactions produced by the dyes between all drug hypersensitivity is still undefined. Bhatia et al. reported a case study where about 2210 patients were exposed to tartrazine consisting medicines, 3.8% of patients resulted in adverse drug reactions, and symptoms lessened within 2 days after terminating the therapy. On the other hand, medicines that lack tartrazine-based compound do not produce any adverse drug reactions. Moreover, the oral iron supplement consisting of various dyes in formulation when given orally, a 43-year-old woman had several adverse reactions like facial erythema, skin swelling, and itching after 24 h of administration of iron supplement. Similarly, yellow dye tartrazine is responsible for a possible reason for worsening allergic rhinitis, urticaria, and asthma in atopic individuals. Interestingly, a double-blind, placebo-controlled, crossed-over challenge with 35 mg of tartrazine was performed in 26 atopic patients. There were no notable cutaneous, respiratory, or cardiovascular effects compared to placebo (Barbaud, 2014).

19.5.6.2 Povidone

Povidone is widely known as PVP, composed of various synthetic polymers with molecular weight ranging from 10,000 to 70,000 Da similar to those seen in plasma proteins. There is no standardized method available for testing PVP. According to Michavila-Gomez and colleagues, an anaphylactic reaction occurred in a 4-year-old boy after taking an oral prednisolone solution containing povidone. A prick test with pure noniodinated povidone (25 mg/mL) yielded a positive result. However, an oral provocation test with methylprednisolone at 20 mg yielded a negative result (Barbaud, 2014).

19.6 Conclusion

Excipients designed for the adult population may not always be suitable for pediatric populations. Hence, a thorough evaluation of safety for the off-label pediatric formulation must be conducted with extensive planning. Furthermore, to ensure the safety of the excipients used in off-labeled pediatric formulation, clinical trials must perform on the pediatrics to assess the adverse drug reactions and other safety concerns. Special care must be taken while taking "off label" formulation, and advice from doctors and experts is essential to prevent toxic reactions. The regulatory guidelines given by ICH and other regulatory bodies should be followed for the drug and excipients to assure its safety and quality throughout the formulation process. The compatibility study between APIs and excipients should be done comprehensively using novel analytical techniques such as isothermal microcalorimetry and solid-state NMR. Moreover, an extensive investigation should be performed on novel excipients when they come into the pharmaceutical market, including its safety and interaction quantification with an active ingredient.

Abbreviations

API	active pharmaceutical ingredients
CDER	Center for Drug Evaluation and Research
COA	certificate of analysis
GMP	good manufacturing practices
ICH	International Conference on Harmonization
IM	intramuscular
IPEC	International Pharmaceutical Excipients Council
IV	intravenous
MCC	Microcrystalline cellulose
PVP	polyvinylpyrrolidone
PQG	Pharmaceutical Quality Group
US FDA	US Food and Drug Administration
WHO	World Health Organization

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